



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
*SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN***

TECHNICAL MATHEMATICS P1/*TEGNIESE WISKUNDE V1*

MAY/JUNE/*MEI/JUNIE* 2026

MARKING GUIDELINES/*NASIENRIGLYNE*

MARKS/*PUNTE*: 150

Marking Codes/ <i>Nasienkodes</i>	
A	Accuracy/ <i>Akkuraatheid</i>
AO	Answer only/ <i>Slegs antwoord</i>
CA	Consistent Accuracy/ <i>Volgehoue Akkuraatheid</i>
F	Formula/ <i>Formule</i>
M	Method/ <i>Metode</i>
R	Rounding/ <i>Afronding</i>
NPR	No Penalty for Rounding/ <i>Geen Penaliserings vir Afronding nie</i>
NPU	No Penalty for Units omitted/ <i>Geen Penaliserings vir Eenhede Weggelaat nie</i>
S	Simplification/ <i>Vereenvoudiging</i>
SF	Substitution in Correct Formula/ <i>Vervanging in Korrekte Formule</i>

**These marking guidelines consist of 24 pages.
*Hierdie nasienriglyne bestaan uit 24 bladsye.***

NOTE:

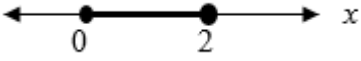
- If a candidate answers a question TWICE, mark only the FIRST attempt.
- If a candidate cancels out an answer and does not attempt the question again, mark the cancelled out work.
- Consistent accuracy (CA) applies in all aspects of the marking guidelines where indicated.

**LET
WEL:**

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Indien 'n kandidaat 'n vraag kansleer en nie weer 'n poging aanwend nie, merk die uitgekansleerde werk.
- Volgehoue akkuraatheid (CA) is deurgaans op alle aspekte van die nasienriglyne van toepassing waar aangedui.

QUESTION/VRAAG 1

1.1.1	$x^2 - 8x = 0$ $x(x - 8) = 0$ OR/OF $x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(0)}}{2(1)}$ $x = 0$ or/of $x = 8$	✓ factors/faktore/SF A ✓ both values of x beide waardes van x CA AO: Full Marks/ Volpunte (2)
1.1.2	$4 + x^2 + 3x = 5$ $x^2 + 3x - 1 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-(3) \pm \sqrt{(3)^2 - 4(1)(-1)}}{2(1)}$ $x \approx -3,30$ or/of $x \approx 0,30$	✓ std. form /vorm A ✓ SF CA ✓ value of x/ waarde v x CA ✓ value of x/ waarde van x CA NPR (4)

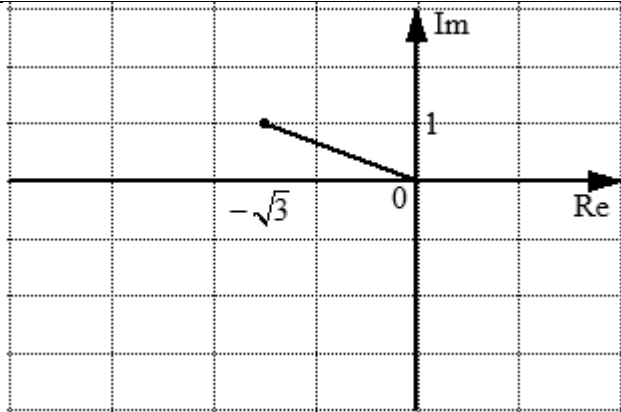
1.1.3	$-x(x - 2) \geq 0$ $0 \leq x \leq 2$ OR/OF $x \in [0; 2]$ OR/OF $x \geq 0$ and / en $x \leq 2$ OR/OF  OR/OF x values from 0 to 2 / x waardes vanaf 0 tot 2	✓ critical values/ kritieke waardes A ✓ correct notation korrekte notasie A OR/OF ✓ critical values/ kritieke waardes A ✓ correct notation korrekte notasie A OR/OF ✓ critical values/ kritieke waardes A ✓ correct notation korrekte notasie A OR/OF ✓ critical values/ kritieke waardes A ✓ correct wording korrekte bewoording A (2)
1.2.1	$y + 2 = 2x$ $y = 2x - 2$	✓ y subject of formula / y onderwerp van formule A (1)

1.4	$\frac{110100_2}{2}$ $= \frac{52}{2} \quad \text{OR/OR} \quad = \frac{110100_2}{10_2}$ $= 26$ $\therefore 26 = 11010_2 \quad = 11010_2$	✓ 52 OR/OR A 10₂ ✓ 11010₂ CA AO: Full marks/ Volpunte No penalty for base 2 omitted/ Geen penalisering vir basis 2 weggelaat (2)
		[22]

QUESTION/VRAAG 3

3.1.1	$(2y^3)^4$ $= 16y^{12}$ OR/ OF $= 2^4 y^{12}$	$\checkmark 16y^{12}$ OR/ OF $2^4 y^{12}$	A (1)
3.1.2	$(3\sqrt{a} - \sqrt{b})(3\sqrt{a} + \sqrt{b})$ $= 9a + 3\sqrt{ab} - 3\sqrt{ab} - b$ $= 9a - b$ OR/ OF $= 9 \times a - b$	$\checkmark$ Expansion/ <i>Uitbreiding</i> $\checkmark$ S AO: Full Marks/ Volpunte	A CA (2)
3.1.3	$\frac{3^{4x+1} \cdot 81^{-x}}{3}$ $= \frac{3^{4x+1} \times 3^{-4x}}{3}$ $= 3^{4x+1-4x-1}$ $= 3^0 = 1$ OR/OF $\frac{3^{4x+1} \cdot 81^{-x}}{3}$ $= \frac{3^{4x} \cdot 3 \times 3^{-4x}}{3}$ $= 1$ OR/OF $\frac{3^{4x+1} \cdot 81^{-x}}{3}$ $= \frac{3^{4x} \cdot 3}{3 \cdot 3^{4x}}$ OR/OF $\frac{81^x \cdot 3}{3 \cdot 81^x}$ OR/OF $\frac{81^x \cdot 3 \times \frac{1}{81^x}}{3}$ $= 1$	$\checkmark 3^{-4x}$ $\checkmark$ exp. prop/ <i>eksp. eienskap</i> $\checkmark$ S OR/OF $\checkmark 3^{4x} \cdot 3$ $\checkmark 3^{-4x}$ $\checkmark$ S OR/OF $\checkmark 3^{4x} \cdot 3$ OR/OF $\frac{81^x \cdot 3}{81^x}$ $\checkmark 3^{4x}$ OR/OF 81^x OR/OF $\frac{1}{81^x}$ $\checkmark$ S AO: 1 Mark/ punt	A CA CA A A CA A CA (3)

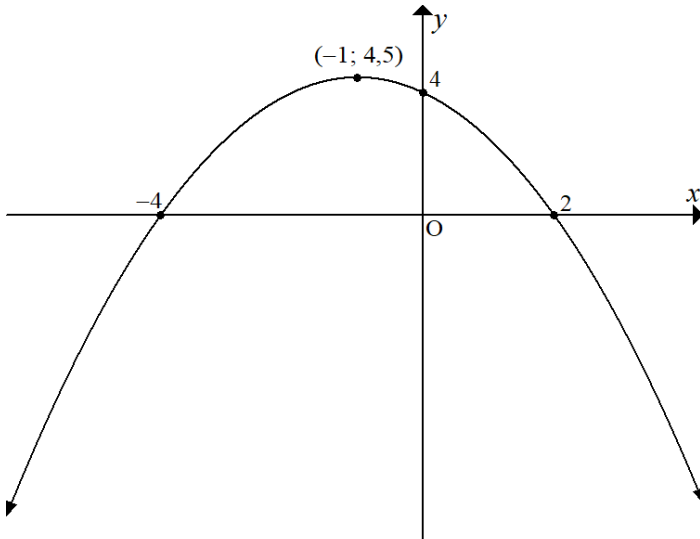
3.1.4	$\log 4 + \log 25 - \log_2 16$ $= \log (4 \times 25) - 4 \log_2 2$ OR/ OF $= \log (2^2 \times 5^2) - \log_2 2^4$ $= \log 100 - 4(1)$ $= 2 \log 10 - 4 \log_2 2$ $= 2 - 4$ $= -2$	✓ log prop./eienskap A ✓ log prop./eienskap A ✓ S CA AO: 1 mark/ punt (3)
3.2	$\log x - \log 5 = -1$ $\log \frac{x}{5} = -1$ OR/ OF $\log \frac{x}{5} = -\log 10$ $\frac{x}{5} = 10^{-1}$ OR/ OF $\log \frac{x}{5} = \log 10^{-1}$ $\frac{x}{5} = \frac{1}{10}$ $\therefore x = \frac{1}{2}$ OR/ OF $\log x - \log 5 = -1$ $\log x - \log 5 + \log 10 = 0$ $\log \frac{10x}{5} = 0$ $\frac{10x}{5} = 10^0$ $10x = 5$ $\therefore x = \frac{1}{2}$	✓ log prop./eienskap A ✓ 10^{-1} OR/ OF A $\log 10^{-1}$ ✓ S CA OR/ OF ✓ log prop./eienskap A ✓ 10^0 A ✓ S CA AO: 1 mark/ punt (3)

3.3.1		✓ quadrant / kwadrant A ✓ co-ordinates / koördinate A NOTE /LET WEL: If only the point is indicated award FULL marks/ Indien slegs die punt aangedui word, ken VOLPUNTE toe (2)
3.3.2	$z = -\sqrt{3} + i$ # $r = \sqrt{(-\sqrt{3})^2 + (1)^2}$ $= 2$ ref / verw. $\angle = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ OR/OF $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ OR/OF $\sin^{-1}\left(\frac{1}{2}\right)$ $= 30^\circ$ $\theta = 180^\circ - 30^\circ = 150^\circ$ $\therefore z = 2(\cos 150^\circ + i \sin 150^\circ)$ OR/OF $2\text{cis}150^\circ$ OR/OF $2 \angle 150^\circ$	✓ SF A ✓ modulus CA ✓ ref. angle/ verw. hoek A ✓ θ in correct quadrant/ θ in korrekte kwadrant CA ✓ polar form/ polêre vorm CA AO: 1 mark/ punt (5)
3.4	$m + ni = -3i + 12$ $\therefore m = 12$ and / en $n = -3$	✓ m value/waarde A ✓ n value/waarde A (2)
		[21]

QUESTION/VRAAG 4

4.1.1	$y - \text{intercept/afsnit} = 4$ OR/ OF $(0;4)$	✓ $y = 4$ Accept/Aanvaar : 4	A (1)
4.1.2	$-\frac{1}{2}x^2 - x + 4 = 0$ $x^2 + 2x - 8 = 0$ $(x+4)(x-2) = 0$ OR/ OF $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4\left(-\frac{1}{2}\right)(4)}}{2\left(-\frac{1}{2}\right)}$ $x = -4$ or / of $x = 2$	✓ equate to /gelykstelling aan 0 ✓ factors / faktore/ SF ✓ both values / beide waardes van x	A A CA (3)
4.1.3	$x \in \mathbb{R}$ OR/ OF $x \in (-\infty; \infty)$ OR/ OF All real values of x / <i>Alle reële waardes van x</i>	✓ domain/ definisieversameling	A (1)
4.1.4	$f'(x) = -x - 1 = 0$ $\therefore x = -1$ $\therefore f(-1) = -\frac{1}{2}(-1)^2 - (-1) + 4$ $= 4\frac{1}{2}$ $\therefore \left(-1; 4\frac{1}{2}\right)$ OR/ OF $x = -\frac{b}{2a}$ $x = -\frac{-1}{2\left(-\frac{1}{2}\right)}$ $\therefore x = -1$ $\therefore f(-1) = -\frac{1}{2}(-1)^2 - (-1) + 4 = 4\frac{1}{2}$ $\therefore \left(-1; 4\frac{1}{2}\right)$	✓ derivative equal to 0/ afgeleide gelyk aan 0 ✓ $x = -1$ ✓ y-value/waarde OR/ OF ✓ SF ✓ $x = -1$ ✓ y-value/waarde	A CA CA A CA CA

	OR/ OF	OR/ OF
	$x = \frac{x_1 + x_2}{2} = \frac{-4 + 2}{2} = -1$ $y = \frac{4ac - b^2}{4a} = \frac{4\left(-\frac{1}{2}\right)(4) - (-1)^2}{4\left(-\frac{1}{2}\right)} = 4\frac{1}{2}$ $\therefore \left(-1; 4\frac{1}{2}\right)$	✓ SF CA ✓ $x = -1$ CA ✓ y-value/waarde CA AO: Full Marks/ Volpunte (3)

4.1.5		✓ shape / vorm A ✓ x- and/ en y-intercepts/ afsnitte CA ✓ turning point/ draaipunt CA (3)
4.1.6	New equation / Nuwe vergelying : $p(x) = -\frac{1}{2}x^2 - x + 7$	✓ new equation / nuwe vergelyking A (1)

4.2.1	$C(0; 4)$	✓ $x = 0$ ✓ $y = 4$	A A (2)
4.2.2	$A(-4; 0)$ & $B(4; 0)$ $\therefore AB = 8$ units/ <i>eenhede</i>	✓ coordinates of A and B/ <i>koördinate van A en B</i> ✓ length of AB/ <i>lengte van AB</i> AO: Full marks/ Volpunte	CA CA (2)
4.2.3	$0 \leq y \leq 4$ OR/ OF $y \in [0; 4]$ OR/ OF $y \geq 0$ and/ <i>en</i> $y \leq 4$ OR/ OF <i>y values from 0 to 4/</i> <i>y waardes van 0 tot 4</i>	✓ critical values / <i>kritieke waardes</i> ✓ correct notation / <i>korrek notasie</i>	CA A (2)
4.2.4	Subst / <i>Vervang</i> $C(0; 4)$: $4 = 2^0 + q$ $\therefore 4 = 1 + q$ $\therefore q = 3$	✓ substitute C / <i>vervang C</i> ✓ value of / <i>waarde van q</i> AO: Full Marks/ Volpunte	CA CA (2)
4.2.5	$m_{BC} = \frac{4-0}{0-4} = -1$ # $\therefore m_{\text{new line/nuwe lyn}} = -1$ $\therefore$ subs. / <i>vervang</i> $A(-4; 0)$: $y - y_1 = m(x - x_1)$ OR/ OF $y = mx + c$ $y - 0 = -1(x + 4)$ $0 = -1(-4) + c$ $-4 = c$ $\therefore y = -x - 4$	✓ gradient/ <i>gradiënt</i> ✓ gradient of parallel line/ <i>gradiënt van ewewydige lyn</i> ✓ sub point A./ <i>vervang punt A</i> ✓ value of c/ <i>waarde van c</i> AO: Full Marks/ Volpunte	CA CA CA (4)
			[24]

QUESTION/VRAAG 5 (Zero marks if wrong Formula is used in this Question / Nul punte indien verkeerde formule in hierdie vraag gebruik word)

5.1.1	10 years / jaar	✓ 10	A (1)
5.1.2	$A = P(1+i)^n$ $= 107\,200(1+7,5\%)^{10}$ $\approx 220\,942$ OR/ OF $\approx 220\,943$ people / mense	✓ F ✓ SF ✓ S R	A CA CA (3)
5.2.1	R504 000	✓ 504 000	A (1)
5.2.2	$A = P(1-i)^n$ $504\,000 = 2\,520\,000(1-0,21)^n$ $\frac{1}{5} = (0,79)^n$ $n = \log_{0,79} \frac{1}{5}$ OR/ OF $n = \frac{\log \frac{1}{5}}{\log 0,79}$ $n = 6,827...$ $n \approx 7$ years / jaar	✓ F ✓ SF ✓ S ✓ log form / vorm ✓ 6,83 Accept 7 yrs./Aanvaar 7yr. NPR	A CA CA CA CA (5)

5.3 #	Amount at the end of 1 year / Bedrag aan die einde v 1 jaar: $A = R\ 60\ 000 \left(1 + \frac{11\%}{4}\right)^{1 \times 4} + R\ 15\ 000$ $\approx R\ 81\ 877,27...$ Amount at the end of 4 years / Bedrag aan die einde v 4 jaar $A = [81\ 877,27...] \times \left(1 + \frac{13,7\%}{2}\right)^{3 \times 2} + R\ 20\ 000$ $A \approx R\ 141\ 845,794...$ Amount at the end of 7 years / Bedrag aan die einde v 7 jaar $A = 141\ 845,794... \times \left(1 + \frac{13,7\%}{2}\right)^{3 \times 2}$ $A \approx R\ 211\ 088,04$ Phumza will reach her goal of saving ENOUGH money for her children's studies / <i>Phumza sal haar doel bereik om GENOEG geld te spaar vir haar kinders se studies</i> OR/ OF $\left\{ \left[60\ 000 \left(1 + \frac{0,11}{4}\right)^4 + 15\ 000 \right] \left(1 + \frac{0,137}{2}\right)^6 + 20\ 000 \right\}$ $\times \left(1 + \frac{0,137}{2}\right)^6$ $A \approx R\ 211\ 088,04$ Phumza will reach her goal of saving ENOUGH money for her children's studies / <i>Phumza sal haar doel bereik om GENOEG geld te spaar vir haar kinders se studies</i> OR/ OF	✓ SF A ✓ 81 877,27... CA ✓ rate A ✓ 141 845,794... CA ✓ R 211 088,04 CA ✓ conclusion/ gevolgtrekking CA OR/ OF ✓ $60\ 000 \left(1 + \frac{0,11}{4}\right)^4$ A ✓ M A ✓ $\left(1 + \frac{0,137}{2}\right)^6$ A ✓ M A ✓ $A \approx R\ 211\ 088,04$ CA ✓ conclusion/ gevolgtrekking CA OR/ OF
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	$A = R\ 60\ 000 \left(1 + \frac{11\%}{4}\right)^{1 \times 4} \times \left(1 + \frac{13,7\%}{2}\right)^{6 \times 2} +$ $R\ 15\ 000 \left(1 + \frac{13,7\%}{2}\right)^{6 \times 2} + R\ 20\ 000 \left(1 + \frac{13,7\%}{2}\right)^{3 \times 2}$ $A \approx R\ 211\ 088,04$ Phumza will reach her goal of saving ENOUGH money for her children's studies / <i>Phumza sal haar doel bereik om GENOEG geld te spaar vir haar kinders se studies</i>	✓ ✓ $\times \left(1 + \frac{13,7\%}{2}\right)^{6 \times 2}$ A ✓ $R\ 15\ 000 \left(1 + \frac{13,7\%}{2}\right)^{6 \times 2}$ A ✓ $R\ 20\ 000 \left(1 + \frac{13,7\%}{2}\right)^{3 \times 2}$ A ✓ $A \approx R\ 211\ 088,04$ ✓ conclusion/ gevolgtrekking CA CA NPR (6)
		[16]

QUESTION/VRAAG 6**Notation will only be penalised in Q 6.1/ Notasie sal slegs in V 6.1 gepenaliseer word**

6.1	$f(x) = -5$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{-5 - (-5)}{h}$ $= \lim_{h \rightarrow 0} \frac{0}{h}$ $\therefore f'(x) = 0$	✓ definition/definisie A ✓ SF A ✓ S CA ✓ 0 CA AO: 0 marks (4)
6.2.1	$y = -3x^{-2} + 4x^{\frac{1}{2}}$ $\frac{dy}{dx} = 6x^{-3} + 2x^{-\frac{1}{2}}$ OR/ OF $\frac{dy}{dx} = \frac{6}{x^3} + \frac{2}{\sqrt{x}}$	✓ $6x^{-3}$ A ✓ $2x^{-\frac{1}{2}}$ A (2)
6.2.2 #	$D_x \left[\left(\frac{3}{x} + 5 \right) (2x - 1) \right]$ OR/ OF $D_x \left[(3x^{-1} + 5)(2x - 1) \right]$ $= D_x \left(6 - \frac{3}{x} + 10x - 5 \right)$ $D_x [6 - 3x^{-1} + 10x - 5]$ $= D_x (1 - 3x^{-1} + 10x)$ $= 3x^{-2} + 10$ OR/ OF $= \frac{3}{x^2} + 10$	✓ $-3x^{-1}$ A ✓ $10x$ A ✓ 1 A ✓ $3x^{-2}$ OR/ OF $\frac{3}{x^2}$ CA ✓ 10 CA ✓ 0 (Implied/ Geïmpliceer) CA (6)
6.3.1	$-8x^{\frac{3}{2}}$ OR/ OF $-8x^{1,5}$	✓ $-8x^{\frac{3}{2}}$ A (1)
6.3.2	$\frac{d}{dx} \left(-8x^{\frac{3}{2}} \right)$ $= -12x^{\frac{1}{2}}$ OR/ OF $-12x^{0,5}$ OR/ OF $-12\sqrt{x}$	✓ $-12x^{\frac{1}{2}}$ CA (1)

6.4.1	$g(-1) = -2(-1)^2 + 3(-1)$ $\therefore g(-1) = -5$	$\checkmark g(-1) = -5$ A (1)
6.4.2	$g(x) = -2x^2 + 3x$ $g'(x) = -4x + 3$ $g'(x) = -4x + 3$ $\therefore g'(-1) = -4(-1) + 3$ $= 7$ $y = mx + c$ OR/OF $y - y_1 = m(x - x_1)$ $-5 = 7(-1) + c$ $y - (-5) = 7(x + 1)$ $c = 2$ $y + 5 = 7x + 7$ $\therefore y = 7x + 2$	$\checkmark$ derivative of /afgeleide van A $\checkmark$ subst. / vervang CA $\checkmark$ gradient/ gradiënt CA $\checkmark$ SF CA $\checkmark$ eqn. of tangent / vergelyking v raaklyn CA (5)
		[20]

QUESTION/VRAAG 7

7.1	$f(x) = (x+3)(x-1)(x-3)$ $= (x^2 - 9)(x-1)$ OR/OF $(x+3)(x^2 - 4x + 3)$ OR/ OF $(x-3)(x^2 + 2x - 3)$ $= x^3 - x^2 - 9x + 9$	✓ expansion / uitbreiding A ✓ standard form / standaardvorm CA (2)
7.2	G(0 ; 9)	✓ x-coordinate / koördinaat A ✓ y-coordinate / koördinaat CA (2)
7.3	Average gradient/ <i>gemiddelde gradiënt</i> = $\frac{9-0}{0+3}$ = 3	✓ SF CA ✓ average gradient/ <i>gemiddelde gradiënt</i> CA (2)
7.4	$f'(x) = 3x^2 - 2x - 9$ $0 = 3x^2 - 2x - 9$ $x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)(-9)}}{2(3)}$ $= \frac{2 \pm \sqrt{112}}{6}$ $\approx -1,43 \text{ or / of } 2,10$ $f(-1,43) = (-1,43)^3 - (-1,43)^2 - 9(-1,43) + 9$ $\approx 16,9$ $f(2,10) = (2,10)^3 - (2,10)^2 - 9(2,10) + 9$ $\approx -5,05$ $\therefore M(-1,43; 16,9) \text{ and / en } N(2,10; -5,05)$ OR/ OF	✓ derivative/ <i>afgeleide</i> CA ✓ equating derivative to 0/ <i>stel afgeleide gelyk aan 0</i> A ✓ SF CA ✓ both values of / <i>beide waardes van x</i> CA ✓ both values of / <i>beide waardes van y</i> CA OR/ OF

	$f'(x) = 3ax^2 + 2bx + c = 0$ $x = \frac{-(2b) \pm \sqrt{(2b)^2 - 4(3a)(c)}}{2(3a)}$ $x = \frac{-(2(-1)) \pm \sqrt{(2(-1))^2 - 4(3(1))(-9)}}{2(3(1))}$ $\approx -1,43 \text{ or / of } 2,10$ $f(-1,43) = (-1,43)^3 - (-1,43)^2 - 9(-1,43) + 9$ $\approx 16,9$ $f(2,10) = (2,10)^3 - (2,10)^2 - 9(2,10) + 9$ $\approx -5,05$ $\therefore M(-1,43;16,9) \text{ and / en } N(2,10;-5,05)$	✓ derivative/afgeleide A ✓ equating derivative to 0/ stel afgeleide gelyk aan 0 A ✓ SF CA ✓ both values of /beide waardes van x CA ✓ both values of / beide waardes van y CA (5)
7.5	$x < -3 \text{ or / of } 1 < x < 3$ OR / OF $x \in (-\infty; -3) \cup (1; 3)$ OR / OF Accept / Aanvaar: x values less than -3 or x values between 1 and 3 / x-waardes kleiner as -3 of x-waardes tussen 1 en 3	✓ $x < -3$: A For / vir: $1 < x < 3$: A ✓ critical values / kritieke waardes A ✓ correct notation / korrek notasie A OR / OF ✓ $x < -3$: A For / vir: $1 < x < 3$: A ✓ critical values / kritieke waardes A ✓ correct notation / korrek notasie A (3)
		[14]

QUESTION/VRAAG 8

8.1	291 KPa	✓ 291 KPa	A (1)
8.2	For maximum/ <i>Vir maksimum</i> $P'(t) = 0$ $P'(t) = -\frac{1}{2}t + 3 = 0 \text{ OR / OF } t = -\frac{b}{2a} = -\frac{(3)}{2\left(-\frac{1}{4}\right)}$ $\therefore t = 6$ $P(6) = -\frac{1}{4}(6)^2 + 3(6) + 291$ $= 300 \text{ KPa}$ OR/OF $Max = \frac{4ac - b^2}{4a}$ $= \frac{4\left(-\frac{1}{4}\right)(291) - (3)^2}{4\left(-\frac{1}{4}\right)}$ $= 300 \text{ KPa}$	✓ derivative/ <i>afgeleide</i> A OR/OF F ✓ equating derivative to 0/ <i>stel afgeleide gelyk aan 0</i> OR/OF SF ✓ $t = 6$ CA ✓ 300 KPa CA OR/OF ✓ F A ✓ SF A ✓ ✓ 300 Kpa CA AO: Full Marks/ Volpunte (4)	
8.3	$-\frac{1}{4}t^2 + 3t + 291 = 100$ $-\frac{1}{4}t^2 + 3t + 191 = 0$ $t = \frac{-(3) \pm \sqrt{(3)^2 - 4\left(-\frac{1}{4}\right)(191)}}{2\left(-\frac{1}{4}\right)}$ $t = -22,28$ or/of $34,28$ $\therefore t = 34,28 \text{ s}$	✓ std. form/ <i>vorm</i> A ✓ SF CA ✓ both values/ <i>beide waardes</i> CA ✓ conclusion/ <i>gevolgtrekking</i> CA AO: Full Marks/ Volpunte NPU (4)	
			[9]

QUESTION/VRAAG 9

9.1.1	$\int (4x^4 + x^{-2} + 5) dx$ $= \frac{4}{5}x^5 - \frac{1}{x} + 5x + C \quad \text{OR/OF} \quad \frac{4}{5}x^5 - x^{-1} + 5x + C$	$\checkmark \frac{4}{5}x^5$ A $\checkmark -\frac{1}{x} \quad \text{OR/OF} \quad -x^{-1}$ A $\checkmark 5x$ A $\checkmark C$ A (4)
9.1.2	$\int \left(\frac{3}{5^{2x}} \right) dx$ $= \int (3 \cdot 5^{-2x}) dx$ $= \frac{3 \cdot 5^{-2x}}{-2 \ln 5} + C \quad \text{OR/OF} \quad = \frac{3}{-2 \cdot 5^{2x} \ln 5} + C$ OR/OF $\int \left(\frac{3}{5^{2x}} \right) dx$ $= \int 3 \left(\frac{1}{5} \right)^{2x} dx$ $= 3 \cdot \frac{\left(\frac{1}{5} \right)^{2x}}{2 \ln \left(\frac{1}{5} \right)} + C$	$\checkmark 3 \cdot 5^{-2x}$ A $\checkmark 3 \cdot \frac{5^{-2x}}{-2 \ln 5} + C \quad \text{OR/OF} \quad \frac{3}{-2 \cdot 5^{2x} \ln 5} + C$ CA OR/OF $\checkmark 3 \left(\frac{1}{5} \right)^{2x}$ A $\checkmark 3 \cdot \frac{\left(\frac{1}{5} \right)^{2x}}{2 \ln \left(\frac{1}{5} \right)} + C$ CA No Penalty if C is omitted/ Geen Penaliserings as C weggelaat is (2)
9.1.3	$\int \left(\frac{x^2 - 10x}{x} \right) dx$ $= \int (x - 10) dx$ $= \frac{x^2}{2} - 10x + C$	$\checkmark S$ A $\checkmark \frac{x^2}{2}$ CA $\checkmark -10x + C$ CA No Penalty if C is omitted Geen Penaliserings as C weggelaat is (3)

9.2	$4 = x + 3$ OR/OF $4 = \frac{5}{x} - 1$ $\therefore x_A = 1$ $0 = \frac{5}{x} - 1$ $\therefore x_C = 5$ $A = \int_1^5 \left(\frac{5}{x} - 1 \right) dx$ $= [5 \ln x - x]_1^5$ $= [5 \ln 5 - 5] - [5 \ln 1 - 1]$ $\approx 4,05 \text{ units}^2 / \text{eenhede}^2$	✓ M A ✓ $x_A = 1$ CA ✓ $x_C = 5$ A ✓ area notation/ <i>area</i> <i>notasie</i> A ✓ integrals/ <i>integreer</i> CA ✓ SF CA ✓ $A = 4,05$ CA (7)
		[16]

TOTAL/TOTAAL: 150